

# Microeconomics I

## Class 4 — Utility maximisation

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7 October 2026

slide 1 of 30

**Slide 1** Title

Class 4 covers MWG 2.D and 3.D: the set of bundles a consumer can afford at prices  $p$  and wealth  $w$ , the best bundle in that set, and the conditions that find it, including when the best bundle contains a zero. Class 3's utility function is what gets maximised today.

## Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

quiz page, not numbered

**Quiz** Start of class, on paper

A short quiz on paper opened the class: circle the letters of your answers in the block on the sheet, write your name on it, and hand in only the strip.

## Plan for today

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|       |                                          |
|-------|------------------------------------------|
| 10:00 | Short quiz                               |
| 10:08 | Practicalities; Ann and Barbara          |
| 10:14 | Budget sets                              |
| 10:26 | The consumer's problem; Walrasian demand |
| 10:50 | Break                                    |
| 11:00 | Kuhn–Tucker: tangency and corners        |
| 11:35 | CES demand; summary                      |

## Slide 2 Plan for today

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The morning at a glance: one problem, maximise  $u$  on a budget set, in three steps. First the set itself and how prices and wealth move it; then the best bundle in it, what makes it exist and the three properties every demand inherits; after the break, the first-order conditions that find it, with and without a zero.

MWG 2.D and 3.D; Muñoz-Garcia chapter 2 is the gentler read.

## Practicalities

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- Problem set 1: due at the start of class Wed 14 Oct, on paper, one copy per group
- Office hours: Wednesdays 12:00–13:00, after class  
up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: [pkocourek.com/teaching/micro-1](http://pkocourek.com/teaching/micro-1) · [pkocourek@ub.edu](mailto:pkocourek@ub.edu)

## Slide 3 Practicalities

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Problem set 1 is the paper set handed out on Tuesday: one copy per group, the answer sheet with all names on top of the working, due at the start of class on Wednesday 14 October.

Office hours are after class on Wednesdays, up to 20 minutes per student; booking by e-mail gives priority. Slides and reading: [pkocourek.com/teaching/micro-1](http://pkocourek.com/teaching/micro-1) · [pkocourek@ub.edu](mailto:pkocourek@ub.edu).

## Ann or Barbara: who is better off?

Same college, similar jobs, both started at \$30 000.

**Ann**

first year: no inflation

**raise 2 %**

**Barbara**

first year: 4 % inflation

**raise 5 %**

## Who is doing better in economic terms? Who is happier?

Shafir, Diamond & Tversky (1997), Quarterly Journal of Economics 112(2), Problem 1.

### Slide 4 Ann or Barbara: who is better off?

Two graduates of the same college, a year apart, similar jobs, the same starting salary. Ann's first year had no inflation and brought a 2 % raise; Barbara's had 4 % inflation and a 5 % raise. Two questions were put to the room: who is doing better in economic terms, and who is happier?

Hold the answer: it is a question about what each salary buys, that is, about budget sets, and homogeneity of degree zero (slide 14) settles it on slide 15. The trap is to read the bigger nominal raise as the bigger raise.

The vignette is Problem 1 of Shafir, Diamond and Tversky (1997), a survey question put to three groups of respondents.

## Most say Ann is better off, and Barbara happier

**better off in economic terms**

**71 %**

said Ann

**happier**

**64 %**

said Barbara

Judged by the number on the payslip, not by what it buys: **money illusion**.

Back to it on slide 15.

Shafir, Diamond & Tversky (1997), "Money Illusion", QJE 112(2): 341–374, Problem 1; three separate groups of US respondents.

### Slide 5 Most say Ann is better off, and Barbara happier

Three separate groups were each asked one question. Asked who is better off in economic terms, 71 % said Ann: they computed in real terms. Asked who is happier, 64 % said Barbara: the nominal 5 % won. Same vignette, two frames, two answers.

Judging by the number on the payslip rather than by what it buys is money illusion. The 64 % do not show that Barbara is better off; they show how people read numbers. These are judgments in a survey, not purchases.

The budget-set argument that says who is right is slide 15.

# Budget sets

MWG 2.D

slide 6 of 30

## Slide 6 Budget sets

Part 1: the set the consumer chooses from, and how prices and wealth move it. MWG 2.D.

### Definition: the Walrasian budget set

$$\mathcal{B}_{p,w} = \{x \in \mathbb{R}_+^2 : p \cdot x \leq w\}$$

prices  $p \gg 0$ , wealth  $w > 0$

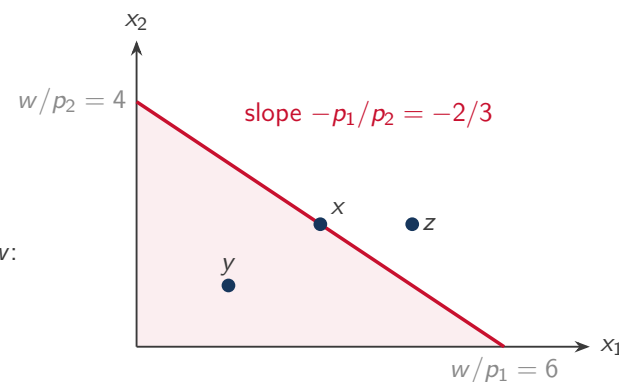
budget line:  $p \cdot x = w$

Which of  $x, y, z$  are in  $\mathcal{B}_{p,w}$ ?

$x$  costs  $12 = w$ ,  $y$  costs  $6$ ,  $z$  costs  $15 > w$ :  
 $x$  and  $y$ .

MWG Definition 2.D.1, Figure 2.D.1.

Here  $p = (2, 3)$ ,  $w = 12$ .



## Slide 7 Definition: the Walrasian budget set

MWG Definition 2.D.1. Read the braces as “the set of all  $x$  in  $\mathbb{R}_+^2$  such that”; the colon is “such that”. A bundle is in  $\mathcal{B}_{p,w}$  when it costs at most  $w$  at the quoted prices, which the consumer takes as given; MWG assumes  $p \gg 0$  and  $w > 0$ .

Costs decide the question on the slide:  $x$  spends exactly  $w$  and is affordable, a bundle on the line is in the set;  $y$  spends less;  $z$  spends more. The intercepts are what one good alone can be bought with,  $w/p_1$  and  $w/p_2$  (not  $p/w$ ). The slope  $-p_1/p_2$  is the market's rate of exchange: giving up one unit of good 1 frees  $p_1$  euros, which buy  $p_1/p_2$  units of good 2.

Two facts used later today:  $p$  is perpendicular to the budget line, and the set is convex, since a mix of two affordable bundles is affordable (slide 8).

## The price vector is perpendicular to the budget line

$x, x'$  on the line:

$$p \cdot (x' - x) = w - w = 0$$

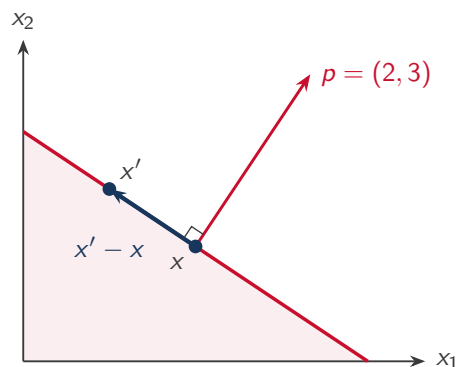
$p$  points out of  $B_{p,w}$ , perpendicular to its edge

$x, x' \in B_{p,w}, \alpha \in [0, 1] \Rightarrow$

$$\alpha x + (1 - \alpha)x' \in B_{p,w}$$

the budget set is convex

MWG 2.D, Figure 2.D.3 and p. 22.



## Slide 8 The price vector is perpendicular to the budget line

Two facts about the shape of  $B_{p,w}$ , both used later today. First,  $p$  is at a right angle to the budget line: two bundles on the line both cost  $w$ , so  $p$  times their difference is zero. The arrow is drawn in price units, so it does not point at a bundle; it points out of the set.

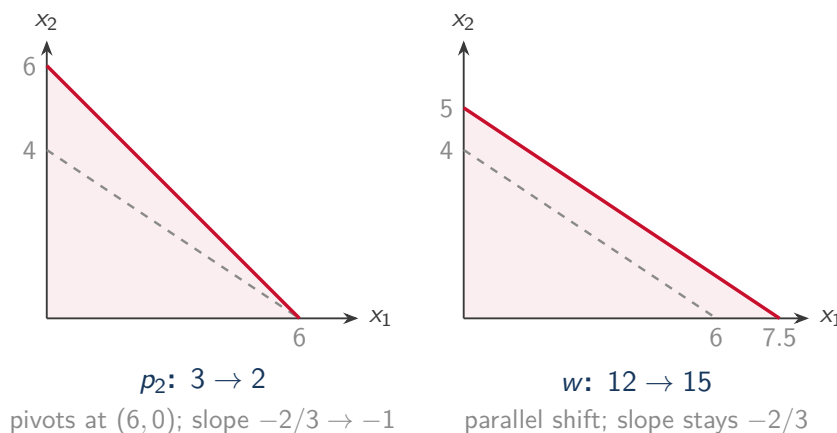
Second, the set is convex: a mix of two affordable bundles is nonnegative and costs at most  $w$ . At an interior optimum the gradient of  $u$  will point along  $p$  (slide 25); convexity of the set is what makes the set of optima convex (slide 17).

An aside: overtime pay, a higher wage after a number of hours, makes a budget set non-convex (MWG Figure 2.D.4).

## Slide 9 A lower price tilts the line; more wealth shifts it

Read the two intercepts and the rest follows. A fall in  $p_2$  leaves  $w/p_1$  where it was and raises  $w/p_2$ : the line pivots at the  $x_1$ -intercept and steepens; the mistake to avoid is moving the  $x_1$ -intercept when only  $p_2$  changed. A rise in  $w$  scales both intercepts by the same factor: a parallel move, same slope, since relative prices did not change.

MWG Figure 2.D.2 is the left panel. Next slide: four changes to work out in pairs.



MWG Figure 2.D.2 (a fall in  $p_2$ ).

## In pairs: what happens to the budget set?

Start:  $p = (2, 3)$ ,  $w = 12$ : intercepts 6 and 4, slope  $-2/3$ .

| new $(p, w)$   | intercepts | slope  | the new set is ...                  |
|----------------|------------|--------|-------------------------------------|
| $(3, 4.5), 18$ | 6 and 4    | $-2/3$ | the same set                        |
| $(2, 3), 9$    | 4.5 and 3  | $-2/3$ | smaller: parallel shift in          |
| $(1, 3), 12$   | 12 and 4   | $-1/3$ | larger: pivot at $(0, 4)$           |
| $(3, 2), 12$   | 4 and 6    | $-3/2$ | neither: they cross at $(2.4, 2.4)$ |

Four minutes. Intercepts first:  $w/p_1$  and  $w/p_2$ .

## Slide 10 In pairs: what happens to the budget set?

Intercepts first,  $w/p_1$  and  $w/p_2$ , then the slope. Row 1 scales every price and wealth by 1.5, so nothing in the inequality changes: this is the homogeneity of slide 14. Row 2 changes  $w$  alone: both intercepts shrink by the same factor, a parallel move inward. Row 3 halves  $p_1$ : only the  $x_1$ -intercept moves, so the line pivots at  $(0, 4)$  and flattens.

Row 4 swaps the prices: one intercept rises, the other falls, and the two lines cross at  $(2.4, 2.4)$ . Each set has bundles the other lacks, so neither is larger; one higher intercept does not make a larger set.

Whether this consumer is better off after a change cannot be read from the sets alone: it depends on her preferences, which is the rest of today.

## Slide 11 The consumer's problem

Part 2: the best bundle in the budget set, why it exists, and the three properties every Walrasian demand inherits. MWG 3.D.

# The consumer's problem

MWG 3.D

## A best affordable bundle exists: continuous $u$ on a compact set

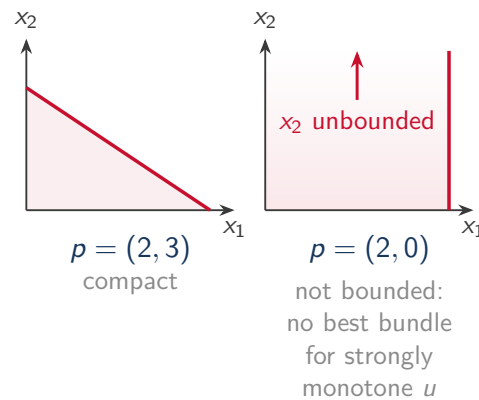
$$\max_{x \in \mathbb{R}_+^2} u(x) \text{ s.t. } p \cdot x \leq w \text{ (UMP)}$$

$\succsim$  rational, continuous, LNS;  $u$  continuous

$p \gg 0$ ,  $u$  continuous  $\Rightarrow$  a solution exists

$B_{p,w}$  closed and bounded ( $x_\ell \leq w/p_\ell$ ): compact

MWG 3.D, Proposition 3.D.1 and its proof, pp. 50–51.



**Slide 12** A best affordable bundle exists: continuous  $u$  on a compact set

The utility maximisation problem picks the best bundle in  $B_{p,w}$ , under MWG's standing assumptions for 3.D:  $\succsim$  rational, continuous and locally non-satiated, so that a continuous  $u$  exists (class 3).

Proposition 3.D.1: with  $p \gg 0$  and  $u$  continuous, a solution exists.

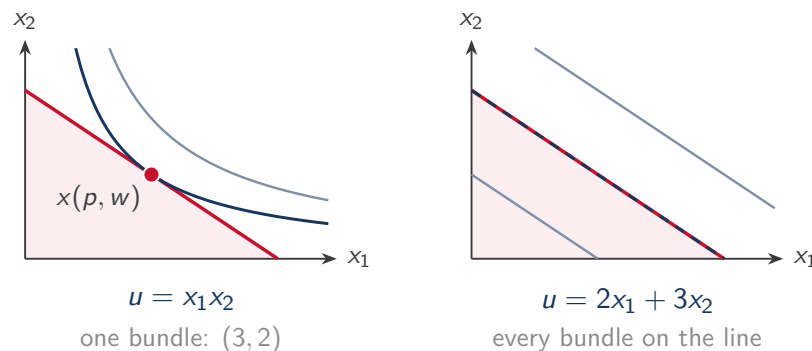
The budget set is closed, because the inequalities are weak, and bounded, because no coordinate can exceed  $w/p_\ell$ : compact. A continuous function attains its maximum on a compact set.

The right panel shows where  $p \gg 0$  enters: with  $p_2 = 0$  good 2 is free, the set is an unbounded strip, and a strongly monotone consumer, or one with  $u = x_1 x_2$ , never stops adding good 2;  $p \gg 0$  is sufficient, not necessary, since with  $u = \min\{x_1, x_2\}$  every bundle  $(6, x_2)$  with  $x_2 \geq 6$  is still best. Neither local non-satiation nor convexity is needed for existence; they are needed later, for Walras' law and for uniqueness.

## Walrasian demand: the best affordable bundles, one or many

$x(p, w)$  = the set of solutions of the UMP: **Walrasian demand**

a function when it is a single bundle at every  $(p, w)$ ; also called Marshallian demand



MWG 3.D, Figure 3.D.1 and footnote 7.

**Slide 13** Walrasian demand: the best affordable bundles, one or many

$x(p, w)$  is defined as a set, the solutions of the problem at  $(p, w)$ , so it is a correspondence; it is a function only when that set has one element. "The" demand bundle need not be unique.

Left: the highest reachable curve of  $u = x_1 x_2$  touches the line at one point. Right: the indifference lines of  $u = 2x_1 + 3x_2$  are parallel to the budget line, so every bundle on the line is best and  $x(p, w)$  is a segment. Which of the two happens is a property of  $\succsim$ : slide 17.

MWG writes Walrasian, not Marshallian, on purpose (footnote 7); either name is fine.

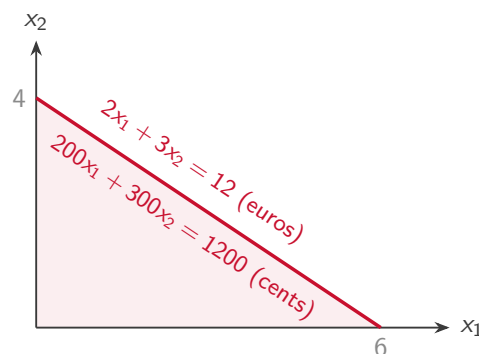
## Scale all prices and wealth by $\alpha > 0$ : demand does not move

$$\{x : \alpha p \cdot x \leq \alpha w\} = \{x : p \cdot x \leq w\}$$

so  $x(\alpha p, \alpha w) = x(p, w)$  for all  $\alpha > 0$

**homogeneous of degree zero**; no assumption on  $u$ : the set is the same

MWG Proposition 3.D.2(i) and its proof, p. 52; the same property is Definition 2.E.1.



**Slide 14** Scale all prices and wealth by  $\alpha > 0$ : demand does not move

Multiply every price and wealth by the same  $\alpha > 0$ , euros to cents say, and the inequality  $\alpha p \cdot x \leq \alpha w$  describes exactly the bundles that  $p \cdot x \leq w$  does. The best bundles in an unchanged set are unchanged: Proposition 3.D.2(i), homogeneity of degree zero.

Nothing about  $u$  or  $\succsim$  was used: this is a property of the set, not an assumption about preferences. Row 1 of the last exercise, where everything was scaled by 1.5, is the same fact. Only relative prices and real wealth matter, which is what the Ann and Barbara question turns on.

**Slide 15** Ann's raise buys about twice as much as Barbara's

## Ann's raise buys about twice as much as Barbara's

Second year, relative to the prices  $p$  and salary  $w$  of the first:

|         | prices        | salary        | budget set                                              |
|---------|---------------|---------------|---------------------------------------------------------|
| Ann     | $\times 1$    | $\times 1.02$ | $\mathcal{B}_{p, 1.02w}$                                |
| Barbara | $\times 1.04$ | $\times 1.05$ | $\mathcal{B}_{1.04p, 1.05w} = \mathcal{B}_{p, 1.0096w}$ |

$\mathcal{B}_{p, 1.0096w} \subset \mathcal{B}_{p, 1.02w}$ : Ann can afford everything Barbara can, and more.

Asked about feelings, people went by the nominal raise, not by what it buys: **money illusion**.

Shafir, Diamond & Tversky (1997), QJE 112(2), Problem 1;  $1.05/1.04 \approx 1.0096$ .

Write each second year in the prices and salary of the first. Ann: prices unchanged, salary  $\times 1.02$ . Barbara: prices  $\times 1.04$ , salary  $\times 1.05$ ; by homogeneity, divide both by 1.04 and her set is the one with prices unchanged and salary  $\times 1.0096$ . In real terms Barbara's raise is 0.96 % and Ann's 2 %, about twice as much, so Ann's budget set contains Barbara's.

The 71 % were right. The 64 % answered a question about feelings with the nominal number: money illusion. Note that  $5 - 4 = 1$  % is close but not the budget-set answer; the exact factor is  $1.05/1.04$ . Whether nominal raises make people happier is psychology; the theory only says what Ann can buy.

Proposition 3.D.2(i) is also a test: a demand that changes when all prices and wealth are scaled by the same factor cannot come from any consumer (row 4 of the next exercise).

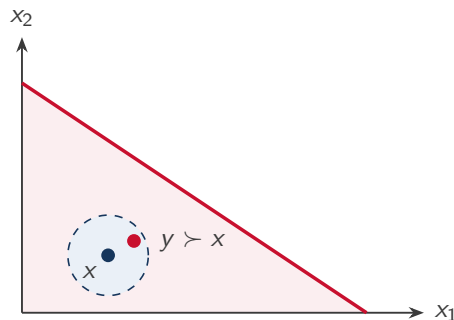
## Walras' law: a locally non-satiated consumer spends all her wealth

$$p \cdot x = w \text{ for every } x \in x(p, w)$$

If  $p \cdot x < w$ :

a small disc around  $x$  is still affordable, and LNS puts a strictly better  $y$  in it

MWG Proposition 3.D.2(ii), Figure 3.D.2. LNS: class 2.



**Slide 16** Walras' law: a locally non-satiated consumer spends all her wealth

Proposition 3.D.2(ii). Suppose a best bundle  $x$  left money unspent. A small enough disc around  $x$  is still affordable, and local non-satiation puts a strictly better bundle inside that disc:  $x$  was not best after all. So every bundle in  $x(p, w)$  costs exactly  $w$ , and from now on the constraint can be written  $p \cdot x = w$ .

This is not an accounting identity true of any consumer; it is what local non-satiation buys, and the reason it is a standing assumption (class 2: LNS is all that Walras' law needs). It fails for a thick indifference band, or for a bliss point inside the budget set: a consumer who wants to be as close as possible to  $(5, 5)$  stops there and leaves the rest unspent when she can afford it.

## Strictly convex preferences: exactly one best bundle

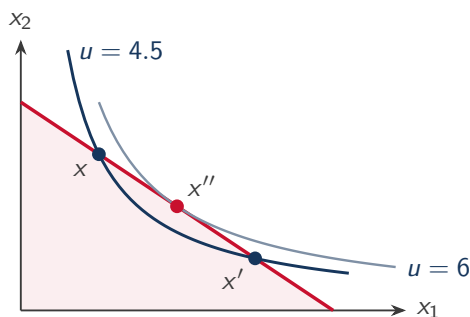
$\succsim$  convex  $\Rightarrow x(p, w)$  is a convex set

$\succsim$  strictly convex  $\Rightarrow x(p, w)$  is one bundle

two best bundles: their midpoint is affordable and strictly better

MWG Proposition 3.D.2(iii), Figure 3.D.3.

Convex  $\succsim \iff$  quasiconcave  $u$ : class 3.



$$x'' = \frac{1}{2}x + \frac{1}{2}x': \text{cost } 12, u = 6 > 4.5$$

**Slide 17** Strictly convex preferences: exactly one best bundle

Proposition 3.D.2(iii). Take two best bundles  $x$  and  $x'$ . Their midpoint is affordable, because the budget set is convex (slide 8), and at least as good as either, because  $\succsim$  is convex, that is,  $u$  is quasiconcave (class 3). So the midpoint is best too, and  $x(p, w)$  is a convex set; slide 13's segment is an example.

If  $\succsim$  is strictly convex the midpoint is strictly better, which contradicts  $x$  being best: at most one best bundle. Convexity alone gives a convex set of optima, not uniqueness. In the picture  $x$  and  $x'$  both give 4.5 while their midpoint gives 6 for the same 12 euros.

Two of the three properties, homogeneity and Walras' law, are tests that any proposed demand must pass: next slide.

## In pairs: could this be a Walrasian demand?

| $x(p, w) =$                                              | homogeneous<br>of degree 0 | Walras' law |                                |
|----------------------------------------------------------|----------------------------|-------------|--------------------------------|
| $\left(\frac{w}{2p_1}, \frac{w}{2p_2}\right)$            | ✓                          | ✓           | Cobb–Douglas: slide 22         |
| $\left(\frac{w}{p_1 + p_2}, \frac{w}{p_1 + p_2}\right)$  | ✓                          | ✓           | $u = \min\{x_1, x_2\}$         |
| $\left(\frac{w}{2p_1}, \frac{w}{3p_2}\right)$            | ✓                          | ✗           | spends $5w/6$                  |
| $\left(\frac{w}{p_1 + 1}, \frac{w}{(p_1 + 1)p_2}\right)$ | ✗                          | ✓           | changes with the unit of money |

Five minutes. Fail either test, and no LNS consumer has this demand.

## Slide 18 In pairs: could this be a Walrasian demand?

Two tests. Homogeneity: replace  $(p, w)$  by  $(\alpha p, \alpha w)$ , prices and wealth together, not prices alone, and see whether  $\alpha$  cancels. Walras' law: compute  $p_1 x_1 + p_2 x_2$  and compare it with  $w$ .

Rows 1 and 2 pass both: half of wealth on each good (Cobb–Douglas, after the break), and equal quantities bought one-for-one (the Leontief preference,  $(p_1 + p_2)x_1 = w$ ). Row 3 is homogeneous but leaves a sixth of wealth unspent, which no locally non-satiated consumer does. Row 4 spends everything but is not homogeneous: the +1 adds one unit of money to a price, so the bundle changes with the unit of money, as the numbers show ( $x_1 = 4$  at  $p = (2, 3)$ ,  $w = 12$ ; 4.5 at  $(3, 4.5)$ , 18).

The logic is Proposition 3.D.2: a continuous  $u$  with local non-satiation implies both properties, so a formula that fails either cannot be the demand of any such consumer, however much it looks like one.

## Slide 19 Break

Ten minutes.

# Break

Back at 11:00

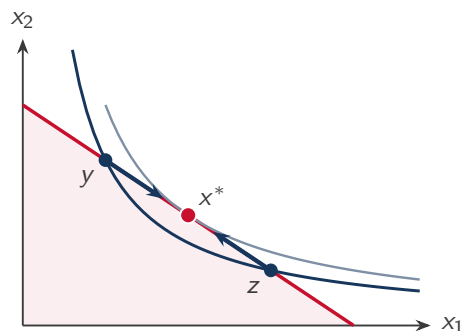
Part 3: how to find the best bundle: tangency where both goods are bought, an inequality where one is not. MWG 3.D.

# First-order conditions

MWG 3.D

slide 20 of 30

## At an interior bundle with $MRS \neq p_1/p_2$ , trading along the line helps



- at  $y$ :  $MRS = 2 > 2/3$ : more  $x_1$
- at  $z$ :  $MRS = 2/9 < 2/3$ : more  $x_2$
- at  $x^*$ :  $MRS = 2/3 = p_1/p_2$

interior optimum:  $MRS_{12}(x^*) = p_1/p_2$

$u = x_1x_2$ ,  $p = (2, 3)$ ,  $w = 12$ ;  $MRS = x_2/x_1$  (class 2).

MWG 3.D, condition (3.D.5) and the argument after it, p. 54.

## Slide 21 At an interior bundle with $MRS \neq p_1/p_2$ , trading along the line helps

Tangency is a consequence, not a rule to memorise. At  $y$  the curve is steeper than the line: the consumer values one unit of good 1 at 2 units of good 2, the market asks only  $2/3$ , so giving up  $2/3$  of good 2 for one of good 1 makes her better off. At  $z$  the comparison runs the other way. Moving along the budget line raises utility until the two slopes agree, and there is the optimum.

MWG's version, condition (3.D.5): if  $u_1/u_2 > p_1/p_2$ , raise  $x_1$  a little and lower  $x_2$  by  $p_1/p_2$  times as much; the budget still holds and utility rises. The argument needs the bundle to be interior, so that both moves are possible; what happens otherwise is slide 23.

## Cobb–Douglas: a constant share of wealth on each good

$u = x_1^\alpha x_2^{1-\alpha}$ ,  $0 < \alpha < 1$ ; maximise  $\alpha \ln x_1 + (1 - \alpha) \ln x_2$  instead same  $\succsim$  (class 3)

$$\text{MRS} = \frac{\alpha x_2}{(1 - \alpha) x_1} = \frac{p_1}{p_2} \Rightarrow \frac{p_1 x_1}{p_2 x_2} = \frac{\alpha}{1 - \alpha}$$

$$\text{with } p \cdot x = w: x_1 = \frac{\alpha w}{p_1}, x_2 = \frac{(1 - \alpha) w}{p_2}$$

$$\alpha = \frac{1}{2}, p = (2, 3), w = 12: x = (3, 2) \text{ slide 21}$$

A share  $\alpha$  of wealth goes to good 1, whatever the prices.

MWG Example 3.D.1 (there  $u = k x_1^\alpha x_2^{1-\alpha}$ ), pp. 55–56.

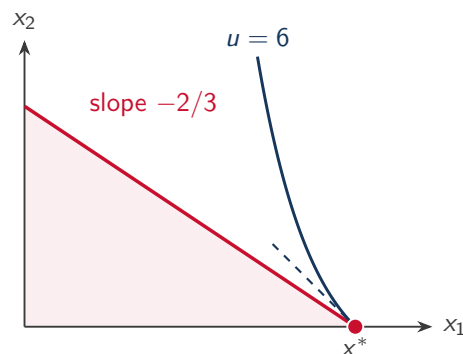
## Slide 22 Cobb–Douglas: a constant share of wealth on each good

Maximising  $\alpha \ln x_1 + (1 - \alpha) \ln x_2$  instead of  $x_1^\alpha x_2^{1-\alpha}$  changes nothing: the log is strictly increasing, so it is the same preference and the same demand (class 3), with easier derivatives. Since  $\ln 0 = -\infty$  the optimum is interior, so both first-order conditions hold with equality, and the budget binds.

Setting the MRS equal to  $p_1/p_2$  (slide 21) gives spending in the ratio of the exponents; the budget then fixes the levels. The result: a share  $\alpha$  of wealth on good 1 whatever the prices, so a rise in  $p_1$  leaves spending on good 2, and hence  $x_2$ , unchanged. With the running numbers, (3, 2) is slide 21's tangency point.

MWG Exercise 3.D.1 asks you to verify Proposition 3.D.2 for this demand; row 1 of the last exercise did two of its three parts, and the third holds because  $\alpha \ln x_1 + (1 - \alpha) \ln x_2$  is strictly concave. How the bundle moves with  $w$  is Tuesday's topic.

## The same recipe can ask for a negative quantity



$$u = x_1 + \ln(1 + x_2)$$

class 3, slide 27;  $p = (2, 3)$ ,  $w = 12$

$$\text{MRS} = 1 + x_2 = \frac{2}{3} \Rightarrow x_2 = -\frac{1}{3}$$

$$\text{best affordable: } x^* = (6, 0)$$

there  $\text{MRS} = 1 > 2/3$ : the curve is steeper than the line

Tangency is a condition for interior bundles only.

MWG Figure 3.D.4(b): a boundary solution.

## Slide 23 The same recipe can ask for a negative quantity

Apply slide 21 blindly to  $u = x_1 + \ln(1 + x_2)$ : the MRS is  $1 + x_2$ , setting it equal to  $2/3$  gives  $x_2 = -1/3$ , and the budget then gives  $x_1 = 6.5$ . That is not a bundle, and reporting it is the mistake to avoid.

What went wrong: at every affordable bundle the consumer values good 1 at more than the market does ( $1 + x_2 \geq 1 > 2/3$ ), so she keeps trading good 2 for good 1 until she has none left. The best bundle is the corner (6, 0), where the curve is still steeper than the line: she would go on, but  $x_2 \geq 0$  binds. This is MWG Figure 3.D.4(b). Tangency is a condition for interior bundles only; the conditions that know about the axes are next.

## Kuhn–Tucker: one set of conditions for interior and corner

$u$  continuously differentiable,  $x^* \in x(p, w) \Rightarrow$  there is  $\lambda \geq 0$  with

$$\frac{\partial u(x^*)}{\partial x_\ell} \leq \lambda p_\ell \text{ for each good } \ell, \text{ with equality if } x_\ell^* > 0 \quad (3.D.1)$$

in vectors:  $\nabla u(x^*) \leq \lambda p$  and  $x^* \cdot [\nabla u(x^*) - \lambda p] = 0$  (3.D.2), (3.D.3)

sufficient too, for  $x^*$  with  $p \cdot x^* = w$ ,

if  $u$  is quasiconcave, monotone,  $\nabla u(x) \neq 0$  for all  $x$  p. 55

MWG 3.D, p. 53; the constraint qualification always holds in the UMP (footnote 8).

## Slide 24 Kuhn–Tucker: one set of conditions for interior and corner

The necessary conditions, MWG (3.D.1), for continuously differentiable  $u$ : if  $x^*$  is best, some  $\lambda \geq 0$  makes every marginal utility at most  $\lambda$  times the price, with equality for every good actually bought. Read them good by good: either  $x_\ell^* = 0$ , or the condition holds with equality. Interior: all equalities,  $\nabla u = \lambda p$ , which is slide 21. Corner: for the good not bought, marginal utility per euro is at most  $\lambda$ , so buying a little of it would not pay. The budget is not in (3.D.1), because Walras' law gives  $p \cdot x^* = w$  at every optimum; to use the conditions as a sufficient test, check it too: for  $u = 2x_1 + 3x_2$ ,  $(1, 1)$  has  $\nabla u = p$  ( $\lambda = 1$ ) but costs  $5 < 12$ .

"Equality if  $x_\ell^* > 0$ " means if, not an equivalence: a good with  $x_\ell^* = 0$  may still satisfy its condition with equality (row 3 of Exercise 3, slide 28). For a bundle with  $p \cdot x^* = w$  the conditions are also sufficient when  $u$  is quasiconcave and monotone with  $\nabla u(x) \neq 0$  for all  $x$ . Without the last requirement, a bad bundle on the line passes: for  $u = (x_1 + x_2 - 3)^3$  and  $w = 7.5$ ,  $(1.5, 1.5)$  costs 7.5 and has  $\nabla u = 0$ , so it passes with  $\lambda = 0$ , yet  $u(3.75, 0) \approx 0.42 > 0$ . The constraint qualification always holds in this problem (footnote 8).

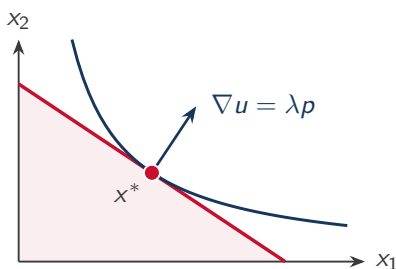
## Slide 25 At a corner the gradient need not point along $p$

Left, both goods bought: the gradient is perpendicular to the indifference curve,  $p$  is perpendicular to the budget line (slide 8), and at a tangency the two directions coincide, so  $\nabla u = \lambda p$ ; here  $\nabla u = (x_2, x_1) = (2, 3) = 1 \cdot p$ .

Right, good 2 not bought: good 1 is bought, so its condition is an equality and fixes  $\lambda = 1/2$ ; for good 2 the marginal utility 1 falls short of  $\lambda p_2 = 3/2$ . The gradient leans toward good 1 relative to  $p$ : she would trade good 2 for good 1 if she had any.  $MRS = p_1/p_2$  holds at interior optima only; at this corner  $MRS > p_1/p_2$ , at a corner on the other axis it is the reverse (slide 27).

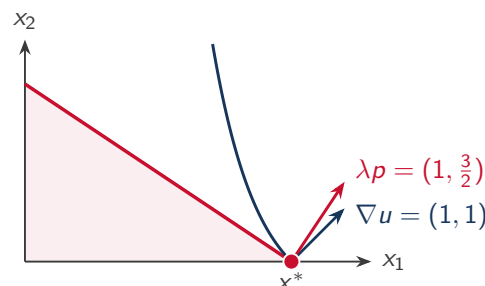
The sufficiency conditions hold here ( $u$  concave, increasing,  $\nabla u \neq 0$ ), so  $(6, 0)$  is the optimum, not just a candidate. The arrow lengths are not comparable across the two panels.

## At a corner the gradient need not point along $p$



$u = x_1 x_2$ , interior

$$\nabla u(x^*) = (2, 3) = \lambda p, \quad \lambda = 1$$



$u = x_1 + \ln(1 + x_2)$ , corner

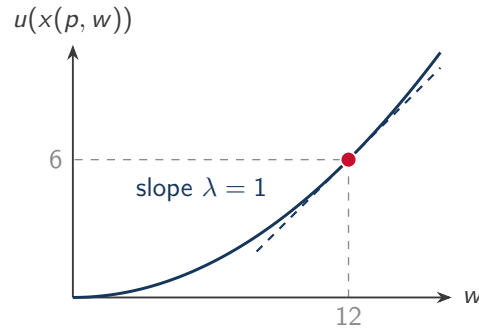
$$x_2^* = 0: \quad \partial u / \partial x_2 = 1 < \lambda p_2 = \frac{3}{2}, \quad \lambda = \frac{1}{2}$$

$\nabla u(x^*) \leq \lambda p$ , with equality in every good that is bought.

MWG Figure 3.D.4.

## $\lambda$ is the marginal utility of wealth, and it is cardinal

| $u$                                         | $\lambda$ at<br>$x^* = (3, 2)$ | $\frac{d}{dw} u(x(p, w))$<br>at $w = 12$ |
|---------------------------------------------|--------------------------------|------------------------------------------|
| $x_1 x_2$                                   | 1                              | $w/12 = 1$                               |
| $\frac{1}{2} \ln x_1 + \frac{1}{2} \ln x_2$ | $1/12$                         | $1/w = 1/12$                             |



$$u = x_1 x_2: u(x(p, w)) = w^2/24$$

Same  $\succsim$ , same bundle,  $\lambda$  rescaled by  $f'(u)$ : like marginal utility (class 3).

MWG 3.D, pp. 54–55;  $p = (2, 3)$ .

## Slide 26 $\lambda$ is the marginal utility of wealth, and it is cardinal

$\lambda$  is the shadow value of the budget: how much the best attainable utility rises per extra euro. With  $u = x_1 x_2$  the best utility at wealth  $w$  is  $(w/4)(w/6) = w^2/24$ , whose slope at  $w = 12$  is 1; and the first-order condition  $x_2 = \lambda p_1$  gives  $\lambda = 1$  too. In general the chain rule and Walras' law give  $\nabla u \cdot D_w x = \lambda p \cdot D_w x = \lambda$  (MWG pp. 54–55).

Write the same preference as  $\frac{1}{2} \ln x_1 + \frac{1}{2} \ln x_2$ : same bundle, but  $\lambda = 1/12$ . Class 3's table said marginal utility does not survive a relabelling  $f \circ u$ , and  $\lambda$  is a marginal utility: it is multiplied by  $f'(u)$ , here  $1/(2u) = 1/12$ . So  $\lambda$  is a property of the representation, not of  $\succsim$ ; the bundle is.

The function  $u(x(p, w))$  in the picture gets a name on Tuesday: the indirect utility function.

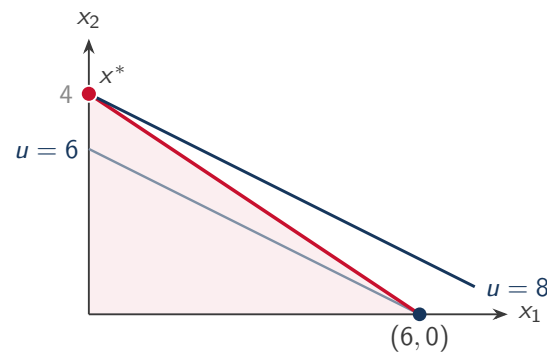
## Perfect substitutes: all on the good with more utility per euro

$$u = ax_1 + bx_2: \text{compare } \frac{a}{p_1} \text{ and } \frac{b}{p_2}$$

$$\frac{a}{p_1} > \frac{b}{p_2}: x = (w/p_1, 0)$$

$$\frac{a}{p_1} < \frac{b}{p_2}: x = (0, w/p_2)$$

equal: the whole budget line slide 13



Here  $u = x_1 + 2x_2$ :  $\frac{1}{2} < \frac{2}{3}$ , so  $x^* = (0, 4)$ ; Kuhn–Tucker:  $\lambda = \frac{2}{3}$ , and  $1 < \frac{4}{3}$  for good 1.

## Slide 27 Perfect substitutes: all on the good with more utility per euro

With linear  $u$  the MRS is the constant  $a/b$ , so it equals  $p_1/p_2$  only by coincidence, and looking for a tangency on a straight line is the wrong move. Compare utility per euro instead: here good 2 gives  $2/3$  per euro and good 1 gives  $1/2$ , so all wealth goes to good 2,  $u = 8$  against 6 at the other corner.

Kuhn–Tucker confirms it: the good bought fixes  $\lambda = 2/3$ , and for good 1 the marginal utility 1 is below  $\lambda p_1 = 4/3$ . This corner lies on the other axis than slide 25's: both happen. When  $a/p_1 = b/p_2$  the indifference lines lie on the budget line and the whole segment is best, slide 13 (right): a convex set of optima, as Proposition 3.D.2(iii) allows for a convex but not strictly convex  $\succsim$ .

## In pairs: Kuhn–Tucker for $u = x_1 + \ln(1 + x_2)$

Find  $x(p, w)$  and  $\lambda$ ; which condition is a strict inequality?

| $p$    | $w$ | $x(p, w)$ | $\lambda$ | strict inequality        |
|--------|-----|-----------|-----------|--------------------------|
| (3, 1) | 12  | (10/3, 2) | 1/3       | none: interior           |
| (3, 1) | 1   | (0, 1)    | 1/2       | good 1: $1 < 3/2$        |
| (1, 1) | 4   | (4, 0)    | 1         | none, although $x_2 = 0$ |

Six minutes. Try interior first; a negative quantity means: set it to zero, check the inequality.

Row 3: a zero and an equality at once: “equality if  $x_\ell > 0$ ” means if, not only if.

## Slide 28 In pairs: Kuhn–Tucker for $u = x_1 + \ln(1 + x_2)$

The recipe: try an interior solution first. The two equalities  $1 = \lambda p_1$  and  $1/(1 + x_2) = \lambda p_2$  give  $x_2 = p_1/p_2 - 1$ , and the budget gives  $x_1$ . If a quantity comes out negative, set it to zero, recompute  $\lambda$  from the good that is bought, and check the inequality for the other.

Row 1 is interior:  $x_2 = 2$ ,  $x_1 = 10/3$ ,  $\lambda = 1/3$ , both equalities. Row 2: the same  $x_2 = 2$  costs more than  $w = 1$ , so  $x_1$  would be negative, which the UMP’s consumption set  $\mathbb{R}_+^2$  rules out (class 3 wrote the quasilinear family on  $\mathbb{R} \times \mathbb{R}_+$ ); with  $x_1 = 0$  all wealth buys  $x_2 = 1$ , good 2 gives  $\lambda = 1/2$ , and good 1 is strictly below:  $1 < 3/2$ . Row 3:  $x_2 = 0$  exactly,  $x_1 = 4$ ,  $\lambda = 1$  from good 1, and good 2 satisfies its condition with equality although nothing of it is bought. A zero and an equality at once: the mistake is to hunt for a strict inequality there.

Since  $u$  is concave and increasing with  $\nabla u \neq 0$ , these are the optima, not just candidates. Note that in row 1  $x_2$  does not depend on  $w$ : extra wealth all goes to good 1, a wealth effect for Tuesday.

## Slide 29 CES demand: the lower $\sigma$ , the more balanced the bundle

Class 3’s one formula returns with its demand. The interior condition  $MRS = (x_1/x_2)^{\rho-1} = p_1/p_2$  gives  $x_1/x_2 = (p_1/p_2)^{-\sigma}$ , and the budget gives the formula on the slide; the weights  $\frac{1}{2}$  cancel. The four points are the same 12 euros spent at  $\sigma = 0, 1, 2$  and  $\infty$ : Leontief buys equal amounts (row 2 of Exercise 2), Cobb–Douglas is slide 22’s (3, 2), and as  $\sigma$  grows the bundle slides toward the cheaper good until, at perfect substitutes, it is the corner of slide 27.

For  $\rho < 1$  the optimum is interior whatever the prices: marginal utility at an axis is infinite (or  $u$  is zero there), so corners need  $\rho = 1$ . The budget share of good 1,  $p_1^{1-\sigma}/(p_1^{1-\sigma} + p_2^{1-\sigma})$ , is constant only at  $\sigma = 1$ .

## CES demand: the lower $\sigma$ , the more balanced the bundle

$$u = \left(\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho\right)^{1/\rho}, \quad \sigma = \frac{1}{1-\rho}$$

class 3, slide 26

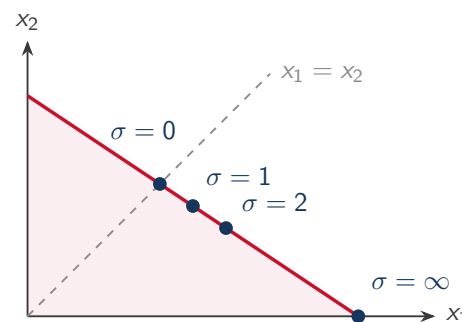
$$x_\ell(p, w) = \frac{w p_\ell^{-\sigma}}{p_1^{1-\sigma} + p_2^{1-\sigma}}$$

$\sigma = 1$ :  $w/(2p_\ell)$ , Cobb–Douglas

$\sigma = 0$ :  $w/(p_1 + p_2)$ , Leontief

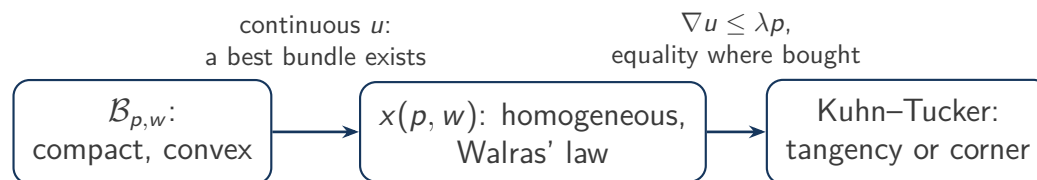
$\sigma \rightarrow \infty$ : all on the cheaper good

$p = (2, 3)$ ,  $w = 12$ ; interior for  $\rho < 1$ , the perfect-substitutes corner at  $\rho = 1$ .



## Today in one line

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Tuesday: indirect utility; wealth and price effects; Roy's identity.

Problem set 1: due at the start of class Wed 14 Oct, one copy per group.

Read MWG 2.E, 3.D and 3.G (or Muñoz-Garcia ch. 2) for Tuesday 13 October.

## Slide 30 Today in one line

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The set is fixed by  $p$  and  $w$ , compact and convex; a continuous  $u$  has a best bundle in it; demand is homogeneous of degree zero (Ann and Barbara), spends all wealth (local non-satiation) and is a single bundle under strict convexity; Kuhn–Tucker finds it, tangency where both goods are bought, an inequality at a corner.

Tuesday 13 October: what the best utility and the demand do when  $p$  and  $w$  change: indirect utility, wealth and price effects, Roy's identity. Reading: MWG 2.E, 3.D and 3.G, or Muñoz-Garcia chapter 2.

Problem set 1 is due at the start of class on Wednesday 14 October, one copy per group.